



Mental workload level assessment based on compounded hysteresis effect

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Abstract

In the domain of neuroergonomics, cognitive workload estimation has taken a significant concern among the researchers. This is because the knowledge gathered from its estimation is useful for distributing tasks among the operators, understanding human capability and intervening operators at times of havoc. Brain signals give a promising prospective for understanding cognitive workload. For this, electroencephalography (EEG) is by far the most efficient modality in interpreting the covert information arising in the brain. The present work explores the feasibility of EEG rhythms for monitoring continuous change occurring in a person's cognitive workload. This continuous monitoring is achieved by graphically interpreting the cumulative effect of changes in EEG rhythms observed in the current instance and the former instance based on the hysteresis effect. In this work, classification is done to predict the data class label using an artificial neural network (ANN) architecture. The proposed model gives a classification accuracy of 98.66%.

Keywords Electroencephalography · Mental workload · Power spectral density · Load graph · Hysteresis effect

Introduction

Estimation of mental workload (MW) is a fast-growing research problem in safety-critical applications. Performance degradation, increase in error count, inattentional deafness, and cognitive tunneling are the significant indices of work overload (Déli and Kisvárdy 2020). Thus, for the smooth functioning of a human-machine system, continuous measurement of MW must be carried out using signature biomarkers (Chen and Epps 2019; Aricò et al. 2014; Orlandi and Brooks 2018). Amongst the several measurement indicators (Liu et al. 2017), information processing of the brain can be easily reflected through the EEG (Amin et al. 2017).

Estimation of cognitive state using EEG can be considered as a method that translates brain signal into time-varying information by extracting features defining different cognitive states (Kohlmorgen et al. 2007; Kothe and

Makeig 2011; Sen et al. 2020). For instance, the logarithmic power for each non-overlapping segmented frequency band was calculated to estimate workload in Baldwin and Penaranda (2012). The authors in Fritz et al. (2014) used combination of power spectrum distribution, eye features and electrodermal activity to estimate MW. Mühl et al. (2014) estimated MW by utilizing the combination of Filter Bank Common Spatial Patterns (FBCSP) and event-related potential (ERP) features. The authors in Amin et al. (2017) measured workload by extracting total and relative sub-band energy from each segment of the EEG signal.

So et al. (2017) evaluated Instant Relative power of EEG frequency to estimate MW. PSD of EEG, ERPs and ECG features are combined to form a fused feature for determining MW in Sen et al. (2020); Zhang et al. (2017b, 2020). Further, the potential of EEG spectral features and band power are investigated in Radüntz (2017) to assess cognitive workload. For efficiently measuring the cross-task mental workload, the sums of PSD from each EEG data segment are used in Ke et al. (2014). The authors in Yu et al. (2015) combined the bilinear common spatial pattern (BCSP), PSD and ERPs of EEG signals to solve the multi-level workload estimation problem. Mahmoud et al. (2017) extracted EEG features using BCSP for successfully

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identifying the cognitive workload. The hybrid functional neuroimaging technique has been studied in Aghajani et al. (2017) to quantify MW. The entropy of wavelet coefficients of EEG are extracted in Zarjam et al. (2013) to estimate cognitive workload. Zhang et al. (2017a) explored the capability of average energy of each frequency band to estimate workload. Spectrograms and periodogram of EEG signals are explored in Çiğdem et al. (2019) and Zammouri et al. (2018), respectively. The authors in Dimitriadis et al. (2015) estimated cognitive workload based on tensorial analysis of EEG. Kim et al. (2018) investigated the effect of task demand transitions from low to high and back to low in a multitasking environment and found significant hysteresis effects on performance. Klaproth et al. (2020) studied perception and responses to alert stimuli using cognitive assistance to ensure they maintain adequate situation awareness. Yi et al. (2019) focused on the presentation rate in the rapid serial visual presentation paradigm to investigate its influence on mental workload, and the separability of brain states during RSVP tasks with different setup of presentation rate.

Recently, deep learning approaches has gained popularity for workload classification. In this regard, Chakladar et al. (2021) have combined the FBCSP method and Long Short-Term Memory (LSTM)-based deep ensemble model for classifying the cognitive state of a user. Further, the authors in Chakladar et al. (2020) have explored a deep hybrid model based on Bidirectional Long Short-Term Memory (BLSTM) and LSTM for the classification of workload levels. Zeng et al. (2018) proposes to use deep convolutional neural networks, and deep residual learning, to predict the mental states of drivers. Yin et al. (2019) proposed a new transfer dynamical auto-encoder in the deep learning framework for recognizing MW levels using neuro-physiological signals.

From the above discussion, it is clear that most of the approaches focus on measuring workload on instance by instance manner rather than in a continuous fashion (Fritz et al. 2014; Zhang et al. 2017b; Mahmoud et al. 2017; Zarjam et al. 2013). Also, the effect of MW measured at former instance is not taken into consideration while measuring the current instance. With this view, the present study aims to investigate the following:

- Defining a feature vector which can classify workload.
- Design and built an ANN model to classify levels of workload.
- Graphical plot of continuous cognitive workload by compounding the load value obtained in the former instance to measure the current instance.

The major contributions of the proposed work are: (i) The work extracts feature based on person specific IAF (Individualized Alpha Frequency) which reduces large inter-

individual variability, (ii) The ANN architecture is trained with data-driven load type rather than pre-defined data tagging which enhances classification accuracy, (iii) Incorporation of leave-one-out method in the ANN model to remove biasness in the training and classification process, and (iv) Combining the discrete output obtained from ANN based on the philosophy of *hysteresis effect* to obtain compounded workload measurement.

The rest of the paper is organized as follows: Section 2 describes the design of the experiment and the experimental protocol. Section 3 presents the feature extraction, the ANN architecture and the approach for MW evaluation. Results obtained from the experiment are discussed in Sect. 4. Section 5 discusses the interpretations of the obtained results. Comparison with the existing literature is done in Sect. 6. Finally, Sect. 7 concludes the work.

Materials

Participants

Ten healthy male volunteers (25 ± 5 years old) were recruited voluntarily, after an explanation of the study, from IIT Kharagpur for the experiment. The experiment procedure followed the principles outlined in the Declaration of Helsinki of 1975, as revised in 2000. Further, the experimental protocol has been approved by the Institutional Ethical Committee (IIT/SRIC/SAO/2017). All participants had normal or corrected-to-normal vision. Participants were currently healthy and reported no history of psychiatric or neurological disease, or medication and drug abuse. They were instructed to avoid alcohol, caffeine, and heavy meals at least four hours before the experiment.

Equipment

The EEG data were recorded at 256 Hz sampling frequency, having 16-bit resolution using a 64-channel scalp electrode cap (Ag/AgCl, RMS, India) following the international 10-20 system. The electrodes corresponding to earlobes are used as reference, and the electrode at the forehead is used as ground. The electrode impedances were kept below $10k\Omega$. From the literature study, it is evident that the frontoparietal, frontal, temporal, parietal and occipital lobes are responsible for MW estimation (Aricò et al. 2014; Ardian et al. 2016; Ke et al. 2014). Following such an observation, in this study, 17 channels (Fp1, Fp2, F3, F4, Fz, F7, F8, T3, T4, T5, P3, P4, Pz, P6, O1, Oz, and O2) were selected for MW evaluation.

Experiment design

The experimental setup and data collection were performed as explained in Samima and Sarma (2019). The Working Memory Test Battery (James Stone 2020) is computer-based cognitive tasks which follow the basic principle of n -back task (Kirchner 1958); thereby providing estimates of short-term and working memory. Workload among the tasks was varied based on the difficulty level (low, medium, high) and speed of inter-element pause duration. The value of ‘ n ’ in the experiment was varied between 3 to 5, where 3 being the lowest and 5 is the highest task difficulty. Each task consists of nine trials with three trials for each difficulty level. The tasks were presented in the following order: Arrow, Matrix, Operation, Reading, Rotation, and Symmetry span. The inter-element pause duration for Arrow span was kept at 600 ms, and the duration was reduced successively by 100 ms for every task. The detailed explanation about the various tasks is available in James Stone (2020).

Experiment

The experiment was carried out in a sound-shielded, air-conditioned laboratory with room temperature maintained at 22 °C at IIT Kharagpur. Two dedicated computers were used for data recording and task activity. During data recording, event markers were used to mark start and end of each task.

The task stimuli were displayed using a Java-based open-source software, *Tatool* (James Stone 2020) onto a 19.5-inch monitor (1600 × 900 resolution) located 70 cm from the participants. Participants completed the questionnaires and consent form before the commencement of the experiment. Each participant’s current alertness and mental state were noted on ‘0’ to ‘10’ scale and Visual Analogue Scale (VAS) which is followed by demonstration phase. Before the experiment, each participant received detailed verbal and visual instructions.

A sample test session was conducted for each participant to minimize the novelty effect. Next, a five minutes of eyes opened resting-state EEG was collected as the baseline, which is followed by the task activities. The experiment protocol is given in Fig. 1. After the session, participants

gave subjective MW rating in the NASA-TLX questionnaire and present mental state on the VAS scale. Also, the subjective rating for task difficulty was recorded. A participant performing task in the experimental environment is shown in Fig. 2.

Method

Assessment of MW being a challenging task, the proposed work is designed to assess the MW of a person involved in continuous monitoring task. The overview of the proposed approach is shown in Fig. 3.

Signal pre-processing

The raw EEG data requires some basic preprocessing steps for better feature extraction. The collected EEG signal passes through a fifth-order Butterworth band-pass filter having cutoff frequencies of 0.5 Hz and 45 Hz. Next, to remove eye blink and muscle movement artifacts, the FORCe algorithm (Daly et al. 2015) was used. This step is followed by data normalization to ensure removal of any unwanted bias. After data cleaning, the next task is to divide them into epochs and then extract features. Using the event marker information, the epochs are extracted from each trial of every task. The duration of each epoch depends on the time taken by each individual to complete each trial. This step is followed by data detrending and baseline removal.

Feature extraction

The general idea of feature vector definition is that the high dimensional input data are transformed into a reduced representation while containing the relevant information from the input data. For this, features are extracted by transforming the preprocessed EEG data into the frequency domain.

In the frequency domain, the EEG spectrum is categorised into electric waves of certain frequencies. The main frequencies are, δ , θ , α , β , and γ . Literature indicates that the peak frequency of EEG power spectrum can vary slightly from subject to subject, and therefore the

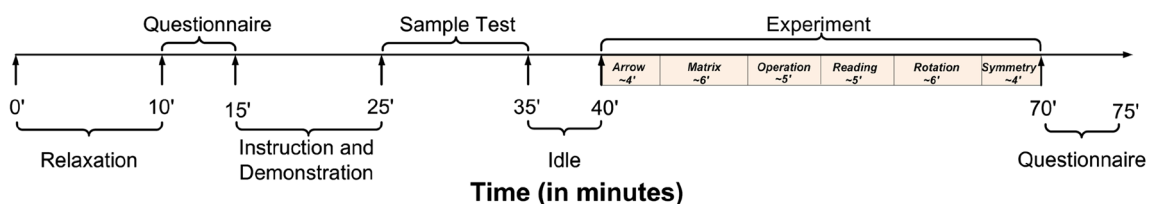


Fig. 1 Description of the experimental procedure

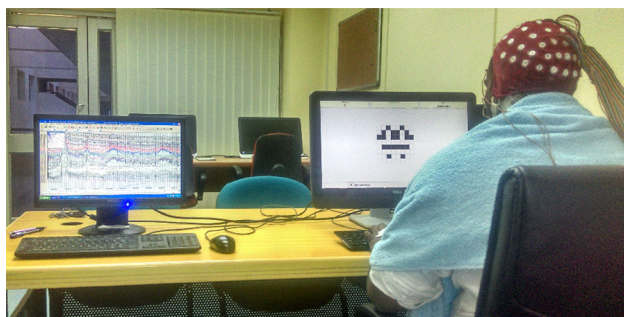


Fig. 2 A representative participant performing the task in the experimental environment

traditional definition of frequency bands could overlook more subtle differences between subjects (Klimesch 1999).

For this, the PSD is calculated using the *IAF*, which is a promising electrophysiological marker of inter-individual differences in cognitive function (Klimesch 1999; Babiloni et al. 2012). As per the definition of *IAF*, EEG rhythms were extracted as $\theta=(IAF-6)$ to $(IAF-4)$, $\alpha_1=(IAF-4)$ to $(IAF-2)$, $\alpha_2=(IAF-2)$ to (IAF) , $\alpha_3=(IAF)$ to $(IAF+2)$, $\beta_1=(IAF+2)$ to 20, $\beta_2=20$ to 30 and $\gamma=30$ to 45 from each epoch of the considered channels. Using these EEG rhythms, a *feature vector* is generated by concatenating the extracted rhythms of each channel. This *feature vector* acts as a representative information for detecting changes in the MW.

In the feature vector, the EEG rhythm elements are taken in the form of ratio between ‘*idle*’ and ‘*experiment*’ conditions. Here, the principle for evaluating cognitive workload considers the recorded ‘*idle*’ data as a reference point for any individual and the actual ‘*experiment*’ data as a measure of their workload. Any workload change

induced during the task performance is evaluated concerning the ‘*idle*’ value. Using this, the feature generation takes place by taking ratio of EEG rhythms during, ‘*experiment*’ to the ‘*idle*’ case.

Among the different rhythms of EEG, the dominant waves are elicited depending on the consciousness of the person. For instance, theta could be observed when a person feels sleepy and becomes constant for wakeful person, on the other hand, gamma rhythms changes sharply when person is actively involved in a task. Similarly, other waves also changes and becomes constant for certain level of consciousness; thus, to cover the minute changes in MW it becomes essential to observe changes in each EEG rhythm. Thus, the feature vector, f considered for evaluating cognitive workload, is defined as:

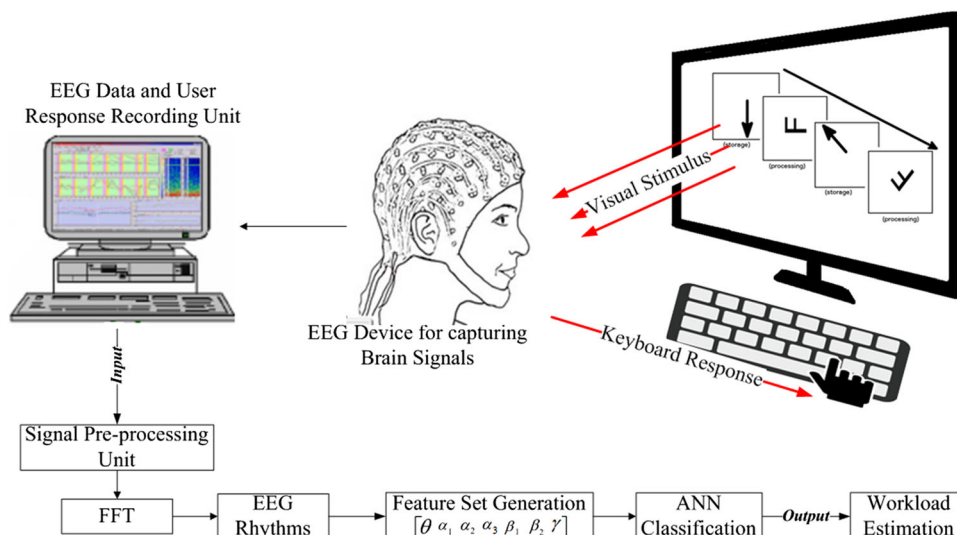
$$f = \left\{ \frac{\theta^{idle}}{\theta^{exp}}, \frac{\alpha_1^{idle}}{\alpha_1^{exp}}, \frac{\alpha_2^{idle}}{\alpha_2^{exp}}, \frac{\alpha_3^{exp}}{\alpha_3^{idle}}, \frac{\beta_1^{exp}}{\beta_1^{idle}}, \frac{\beta_2^{exp}}{\beta_2^{idle}}, \frac{\gamma^{exp}}{\gamma^{idle}} \right\} \quad (1)$$

It is to be noted that in the first three elements, the ratio constitutes ‘*idle*’ versus ‘*exp*’ whereas in the next four elements the ratio is exactly the opposite. This has been done to bring a similar trend for all the feature elements.

ANN model for mental workload estimation

For classification, the proposed methodology uses the ANN model due to its ability to learn from the examples and apply them when a similar event arises. This makes the approach applicable in real time. Moreover, ANN can learn the non linearity present in data and help in accurate prediction of workload. Further, during experimentation, it was observed that ANN outperformed the linear (SVM

Fig. 3 Overview of the proposed approach



with linear kernel) and combination of piecewise linear classifier (Random forest).

Usually, for classification, the features are extracted and tagged based on the load type without considering the actual load-induced (or experienced) on the particular individual. Such approaches assume that load imposed and load-induced are the same for every individual. However, such an assumption may work sometimes but will fail for real-time applications (Ke et al. 2014). This is because every individual experiences a different load under the same load conditions. To address this issue, the proposed work tags the extracted features as ‘low’, ‘medium’ and ‘high’ based on the range of elements of the feature vector.

ANN architecture

The neural network considered for workload classification (see Fig. 4) comprises of *tansig* activation function in the hidden layer due to its efficiency in modelling non-linear problems (Sahin and Koyuncu 2012). Further, a linear activation function is used in the output layer to accelerate the convergence of the parameter during training (Sibi et al. 2013). The loss function (mse) is minimized using a gradient descent.

The network used in the analysis has the following parameters: (a) 40 neurons, (b) 10000 training iterations, and (c) the error goal MSE=0.01. The number of neurons (40) was obtained heuristically by increasing it until the rating of the validation set improved sufficiently (MSE

decreased less than 1%). There were two alternative stopping rules applied during the training of this network: the number of training iterations and the error goal.

Classification of mental workload

The proposed work divides the dataset into two parts, namely training data and test data based on the leave-one-out method. This work considers orderly selection of the test dataset from ten subjects followed by recursive classification process for ten times. The overall classification process is as follows:

- From every epoch of each participant, the range of all elements of the feature vector is recorded.
- For every vector component, the range is divided into three equal parts.
- For every component, mark the lowest range as ‘low’, middle range as ‘medium’ and highest range as ‘high’ cognitive workload.
- Utilize the information of each component to tag entire training dataset.

From the evaluated feature component, the tagged data is fed into the ANN architecture for training. During testing, the type of load (that is, ‘low’, ‘medium’, ‘high’) is decided based on the combined value obtained from all the elements present in the feature vector as computed by the trained ANN classifier. The overview workload classification approach is shown in Fig. 5.

Fig. 4 The ANN architecture

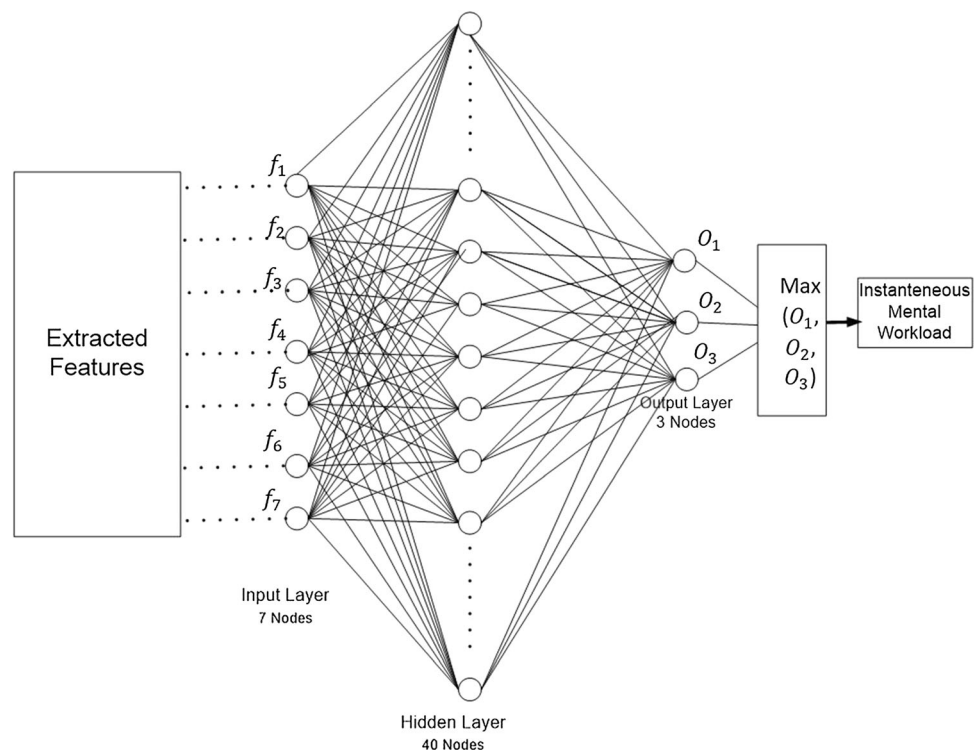
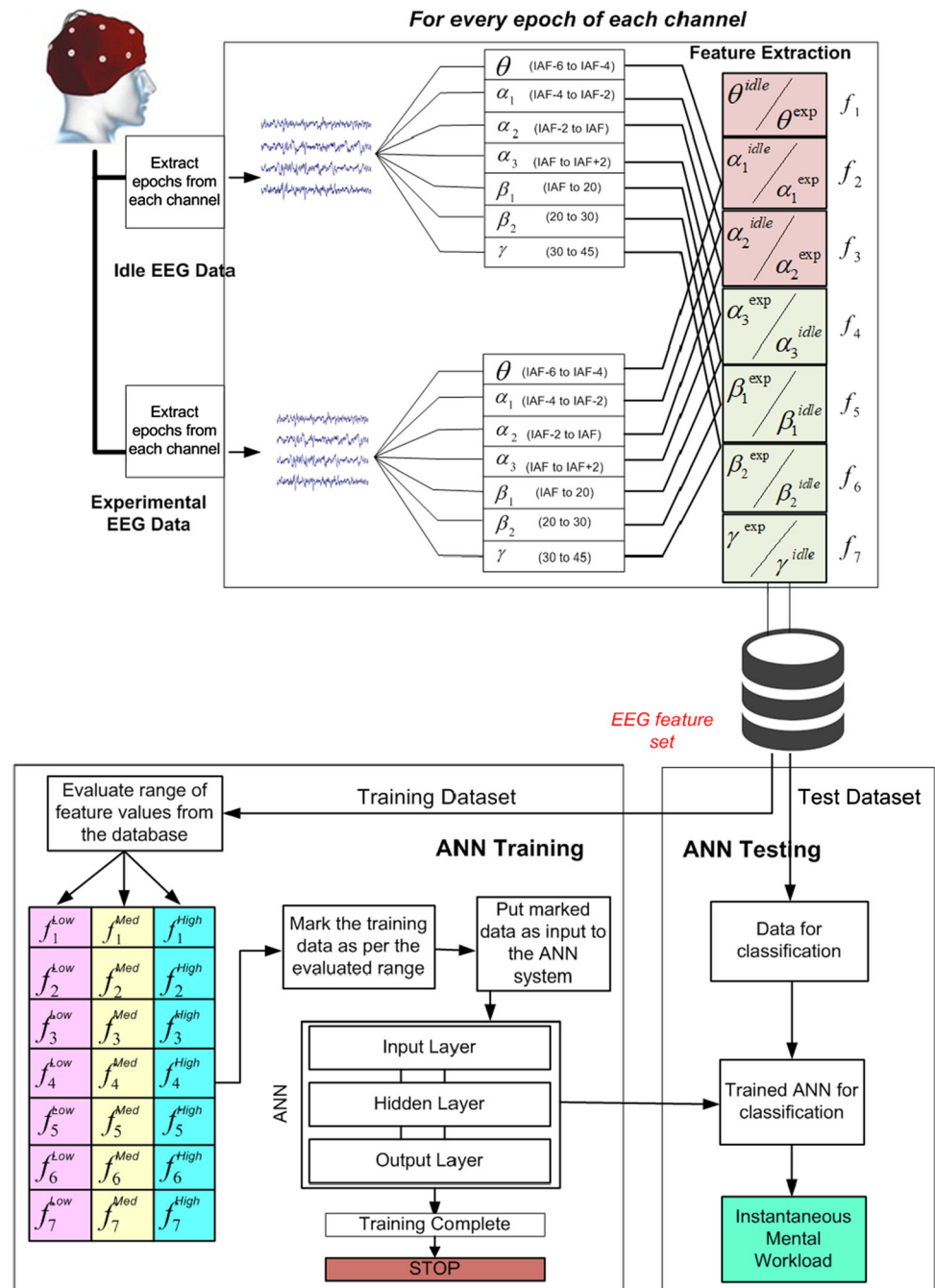


Fig. 5 The working in the ANN architecture



Compounded evaluation of mental workload based on hysteresis Effect

The classification result obtained in Sect. 3.3 gives discrete workload values in an instance by instance manner. From

this, the instantaneous workload is evaluated by measuring the central tendency of the values obtained from each channel. Here, median is considered for evaluating instantaneous workload (refer Eq. 2) as it is not influenced by skewed data.

$$IW = M[f_1, f_2, f_3, f_4 \dots f_c]$$

where f_i = The feature set obtained from i^{th} channel, IW = Instantaneous Workload, M = Median

(2)

Such instantaneous workload values could be misleading at many times, as it may be arising due to more focus on a specific task, or change in concentration. Besides, measuring workload in such a manner emphasized the ‘current’ value of the workload and neglects the effect of workload value obtained at ‘former’ instance. Since workload is a continuous phenomenon, so its evaluation should consider the impact of workload value at the ‘former’ instance to measure the ‘current’ instance, this way, any abruptness and biasness in the workload measurement can be taken care.

The proposed approach utilizes the knowledge gathered from the classification to obtain the compounded workload value. For this, $1/n^{\text{th}}$ factor of instantaneous workload is taken to obtain the contribution of each instantaneous workload. Here, n is the number of continuous instances of high workload in instantaneous observation which may lead to a critical situation. The value of n depends on the criticality of the task being performed. In the proposed work, value of n is taken based on the number of task of same workload being performed, which is 3. This way, for an instance (say t), the workload will be given by Eq. 3 and the compounded workload value is obtained by Eq. 4.

$$IW_t = \frac{M[f_1, f_2, f_3, f_4 \dots f_c]}{n} \quad (3)$$

$$cw_t = \begin{cases} cw_{(t-1)} + IW_t, & \text{if } t > 1 \\ IW_t, & \text{otherwise} \end{cases} \quad (4)$$

where,

cw_t = cognitive workload at t^{th} instance

IW_t = Instantaneous Workload at t^{th} instance

Experimental results

Subjective analysis

Likert scale

For analyzing the differences in VAS states before and after the experiment, the repeated measures ANOVA analysis has been carried out which reveals that the VAS states are significantly different at $p < 0.005$ with $F\text{-value} = 15.21$. Here, the raw scores of VAS rating of each participant has been taken as *response variable* and VAS

observation time has been taken as within subject *factor*. Similarly, by taking raw values of *alertness rating* of each participant as *response variable* and *alertness observation time* as within subject *factor*, Further, the repeated measures ANOVA analysis reveals that the alertness levels are significantly different at $p < 0.001$ with $F\text{-value} = 24.50$.

NASA-TLX

In this study, the NASA-TLX scale has been used to analyze the perceived workload by an individual. The scale focuses on six factors which takes into account all possible aspects interplaying in the perceived workload. One-way ANOVA and Tukey’s pairwise comparison test for multiple mean comparisons has been done for the factors involved in the NASA-TLX scale (refer Fig. 6).

For analyzing the factors involved in the NASA-TLX scale, the *factor response scores* are taken as *response variable* and the *factor type* are considered as *factor* in the one-way ANOVA analysis. From the results, it is evident that the factors are significant at $p < 0.01$ with $F\text{-value} = 3.31$. The mean and standard deviation for each of the factors are as follows: mental demand ($\mu = 69.50$, $\sigma = 17.46$), physical demand ($\mu = 42.10$, $\sigma = 27.84$), temporal demand ($\mu = 62.90$, $\sigma = 23.94$), performance ($\mu = 57.30$, $\sigma = 20.95$), effort ($\mu = 65.30$, $\sigma = 20.27$) and frustration ($\mu = 36.10$, $\sigma = 27.96$). Thus, the results imply that the experiment successfully induces load in the participants and clearly evident from Fig. 10.

Task difficulty analysis

In general, every task affects differently for each individual. The box plot for the task difficulty ratings among the participants is shown in Fig. 7. Further, by taking *task difficulty rating* as *response variable* and *task* as within subject *factor*, the repeated measures ANOVA analysis reveals that the tasks are significantly different at $p < 0.001$ with $F\text{-value} = 11.61$. This low $p\text{-value}$ indicates that ‘means’ for all six group of tasks are not equal which accounts for the variability between the participants.

Behavioural analysis

The behavioural data of each participant was recorded during the task. For each task, the accuracy and reaction

Fig. 6 Difference mean plot for NASA-TLX questionnaire

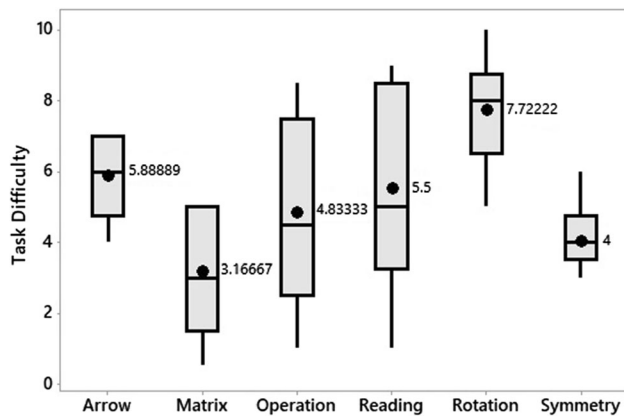
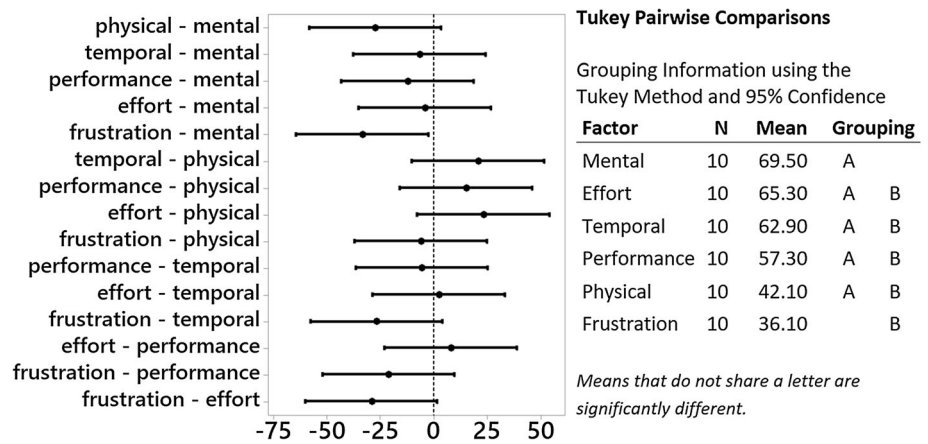


Fig. 7 Self-report of task difficulty

time are evaluated for every task difficulty level. The results of one-way ANOVA analysis show that the accuracy of the tasks are significant at $p < 0.001$ having $F\text{-value} = 14.39$. On the other hand, the response times are found to be significant at $p < 0.001$ having $F\text{-value} = 15.79$. The summary of the accuracy and reaction times are tabulated in Table 1.

From Table 1, it can be observed that the participants' performance degrades at high workload conditions compared to the lower ones. Besides, with an increase in load, the response time increases. The accuracy and response time variation for the tasks is graphically shown in Fig. 8 and 9. In these figures, 1, 2, 3 represent low, medium, and high difficulty level, respectively.

Cognitive workload analysis

In this work, a feature vector is defined to estimate the workload changes. From Fig. 10, the change in values among each task and also within the levels can be easily visualized. In Fig. 10, the range of PSD values for each task and at every difficulty level is easily observable. The

Table 1 Behavioural responses: accuracy and reaction time

Task	Level	Accuracy (%)		Response Time (sec)	
		Mean	SD	Mean	SD
Arrow	Low	96.3	11.11	14.16	2.3
	Medium	66.67	33.33	20.44	3.77
	High	44.44	33.33	37	11.48
Matrix	Low	100	0	5.9	1.32
	Medium	100	0	7.86	1.76
	High	88.89	16.67	10.34	1.51
Operation	Low	96.3	5.56	17	2.88
	Medium	91.36	15.49	17.72	4.15
	High	86.42	14.46	20.04	5.01
Reading	Low	86.42	19.86	17.74	5.42
	Medium	77.78	26.06	19.11	3.74
	High	66.67	28.33	19.53	5.45
Rotation	Low	66.67	40.82	14.92	2.54
	Medium	51.85	37.68	24.92	5.92
	High	40.74	32.39	35.2	5.3
Symmetry	Low	98.77	3.7	17.28	7.7
	Medium	97.53	4.9	21.26	9.72
	High	97.53	4.9	24.19	11.5

feature vector obtained by concatenating the EEG rhythms helps to classify the load levels in the tasks.

Figure 11, indicates that at low workload conditions, the power value of θ and α is higher as compared to β and γ whereas in medium workload condition, the value of these frequencies are more or less same. However, at high workload conditions β and γ components have very high values.

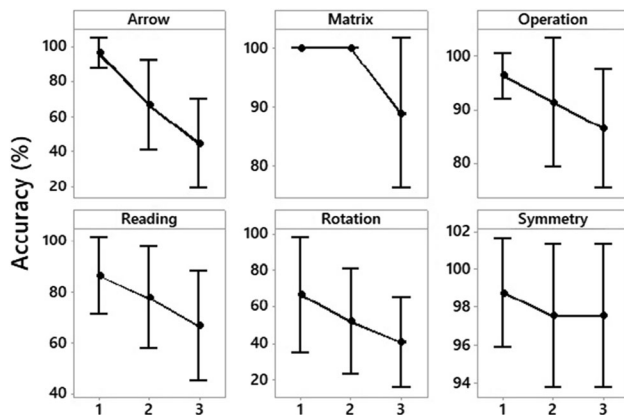


Fig. 8 Interval plot of accuracy for different tasks

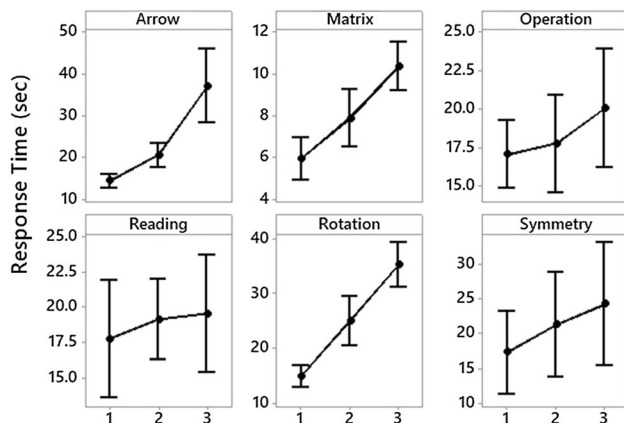


Fig. 9 Interval plot of response time for different tasks

ANN results

The performance of the ANN architecture is measured in terms of *average accuracy*, *Precision*, *Recall*, and *F-score* (Amin et al. 2017). The overall classification accuracy is found to be 98.66%. To analyze the superiority of the proposed approach, popular classification approaches has been implemented, whose classification results are shown in Table 2. The receiver operating characteristics (ROC) curve of the proposed approach is shown in Fig. 12.

Furthermore, CNN and LSTM have been applied on the collected data. In both the architecture we have used the training batch of size $600 \times 30 \times 7$. A 3×3 convolution filter has been used for CNN architecture in all the layers. The architecture consists of three convolution layers where each layer has 15 filters. After each convolution layer, a Pooling layer and a batch normalization layer was used. The output from the last convolution layer is passed into a fully connected layer. The final layer utilizes the soft-max layer whereas all the convolution layers use Rectified Linear Unit (ReLU) activation function.

Similarly for the second architecture, it consists of an input layer, the first sequence-to-sequence LSTM layer, a many-to-one LSTM layer, a 20% dropout layer, and a final sigmoid activation function. The first hidden layer contained 50 LSTM units while the second hidden layer used 10 units. It is observed that, using a dropout of 20% on the input gates and between the final LSTM output and the sigmoid classification layer provided better results than the typically recommended 50% dropout for fully connected layers.

Next, to evaluate the efficiency of these approaches, the experimental data is tested using k-fold classification as done for the proposed ANN approach. The average accuracy obtained for CNN is 95.4% whereas for LSTM model accuracy is 96%. We could observe that the applied approach did not improve the performance of the systems and the network converges to sub-optimal solution due to the lack of samples. Furthermore, it can be observed that the proposed ANN approach is suitable enough to be trained optimally with the limited data. This is in congruence with the real time scenario where the training and MW evaluation is to be performed with limited available data.

Workload graph

The compounded workload value obtained using Eq. 4 helps to understand the trend of the workload variation in a person. This compounded workload variation can be better depicted using a graphical plot (load graph). The load graph of workload variation for every individual performing different task has been plotted in Fig. 13. From the graph, one can easily understand the variation of workload during task performance.

The graph indicates that for every task, the load perceived by each individual is different. Further, one can see that for most of the tasks the workload has increased with increase in difficulty. Empirically, in the graph, it has been considered that the values between '0' to '1' indicate *low* workload, '1' to '2' indicate *medium* workload and above '2' indicate *high* workload level.

Validation through functional connectivity analysis

In this research work, besides considering the subjective and performance analysis (refer Sect. 4.1 and Sect. 4.2), functional connectivity analysis has been carried out to understand the variation of load at each difficulty level in the EEG signals. The aim is to substantiate that the collected raw EEG data has load data of different difficulty levels (that is, *low*, *medium*, *high*). In MATLAB version 2014a, the BRAPH software is used to build the functional

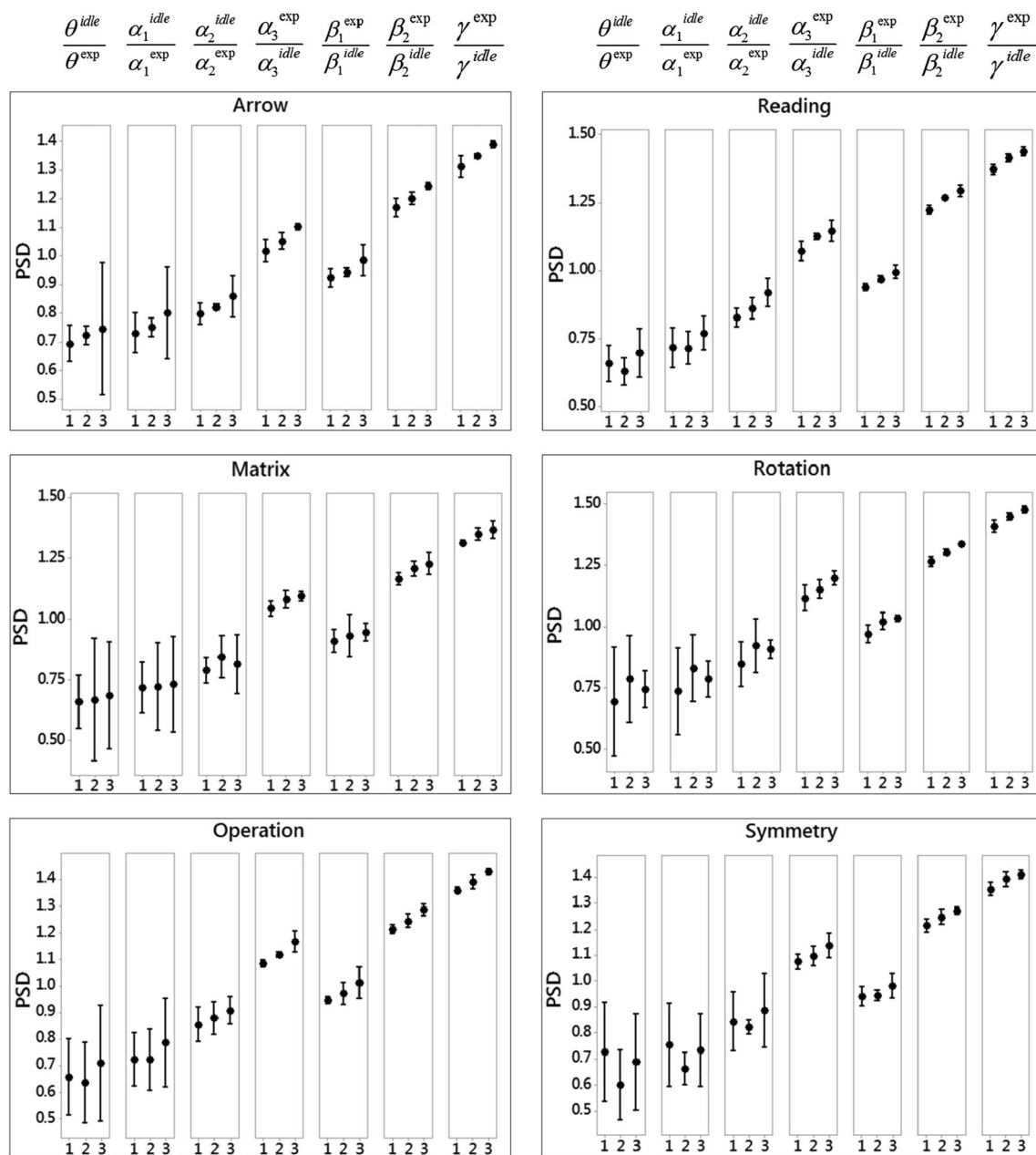


Fig. 10 PSD of EEG rhythms for different tasks. Here, 1,2,3 represent low, medium and high difficulty level, respectively

brain network (Mijalkov et al. 2017). The topological properties of graphs were assessed using global and nodal network measurements (Rubinov and Sporns 2010). The global and nodal measures considered in this study are: *local efficiency*, *clustering coefficient* and *strength*.

The parameter, *clustering coefficient*, is known to have inversely proportional relation to MW (Sciaraffa et al. 2017). From Table 3, it can be observed that for the alpha band of parietal area although the main effect of difficulty level did not reach statistical significance ($p=0.557$), the clustering coefficient values show decreasing trend with increase in workload from ‘low’ to ‘high’ conditions.

Moreover, for the beta band of parietal area, statistical significance is found at $F\text{-value}=95.88$ and $p<0.01$ from ‘low’ to ‘high’ conditions. Further, clustering coefficient values show a decreasing trend with the increase in workload. Further, average *strength* is calculated for the considered brain areas in theta and alpha band. Although the results in Table 4 showed an increasing trend from low to high difficulty levels for both the bands, but, the results are significant for alpha band over parietal and temporal regions (Table 5). Besides, Fig. 14 shows the *local efficiency* in each brain area for the θ and α bands. It can be observed that for both the theta and alpha bands, there are

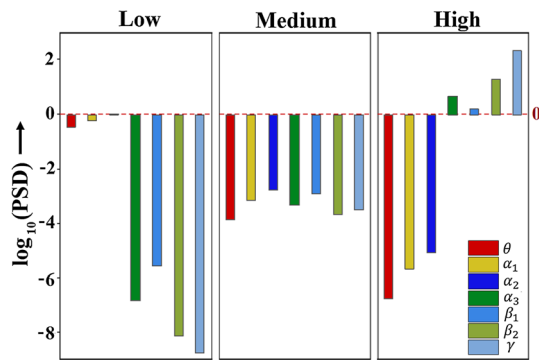


Fig. 11 Change of PSD for each task difficulty level

Table 2 Classification of MW using different approaches

Approach	Average	Precision	Recall	F-Score
Random Forest	97.23	97.13	97.23	97.18
SVM	98.26	98.89	98.26	98.58
ANN	98.66	98.63	98.73	98.68
CNN	95.42	95.42	96.87	96.28
LSTM	96.02	96.02	95.87	95.97

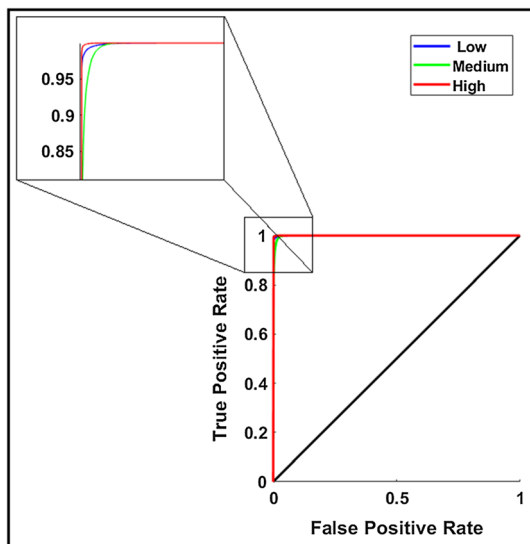


Fig. 12 ROC curve representing classification accuracy

higher values in the low difficulty condition than in the high condition. However, one-way ANOVA result reveals that such a trend is significant over the parietal, occipital, and temporal regions for the theta band but are not significant for the alpha band (Table 6). Figure 15 describes the overall functional connectivity at different difficulty levels. From this, one can observe that with an increase in load, the connectivity decreases and segregates to a smaller area.

Discussion

This work proposes a graphical representation of compounded MW evaluation for analyzing the workload variation among the participants. To meet the criteria, this work proposes a cognitive workload graph, which is dependent on the cognitive workload of the ‘former’ instance to measure the ‘current’ instance. From the analysis of the subjective scale, it can be observed that there is a substantial change in the mood of the participants before and after the experiment. Further, the alertness of the participants has also dropped down towards the end of the experiment. The fact is quite evident from the NASA-TLX results, which indicates that the factor, *mental demand* has been significantly affected for carrying out the experiment. The behavioural analysis also gives enough evidence to support the aforesaid argument. From Table 1, it can be observed that as the difficulty level within the task increases, there is a gradual decrease in accuracy and increase in the response time among the participants. From this observation, it is clear that with an increase in load, there is a substantial change in behaviour among the participants.

The variation of EEG rhythms within the vector for each load level is quite prominent from Fig. 11. This helps to understand the efficiency of approach in classifying the load data. The main focus of the work is to simplify the compounded workload evaluation process through the use of a graphical representation. To fulfill the need, the proposed approach uses the feature vector defined in Eq. 1 and the compounded workload estimation equation proposed in Eq. 4. The graphical interpretations given in Fig. 13 clearly shows the variation of load in each individual. Further, it can be observed that the workload has been successfully induced in the participants during the experiment which is also evident from the subject ratings discussed in Sect. 4.1.

To validate the efficacy of the experimental data, assistance is taken from functional connectivity analysis. Parameters like *clustering coefficient*, *strength*, and *local efficiency* are explored to identify the different load types induced in the EEG signals. From the analysis, it is evident that with an increase in workload, there is decrease in clustering coefficient values. Further, in the beta band of parietal area, statistical significance has been found from ‘low’ to ‘high’ conditions (Table 3) which denote that there is impairment in local connectivity in higher workload conditions (Bullmore and Sporns 2009; Sciaraffa et al. 2017). Moreover, it can be observed that the *strength* value increases with an increase in load. The results are significant for alpha band over parietal and temporal regions (Table 5), which is due to higher inter-connections in these regions during a more demanding task (Ginestet et al.

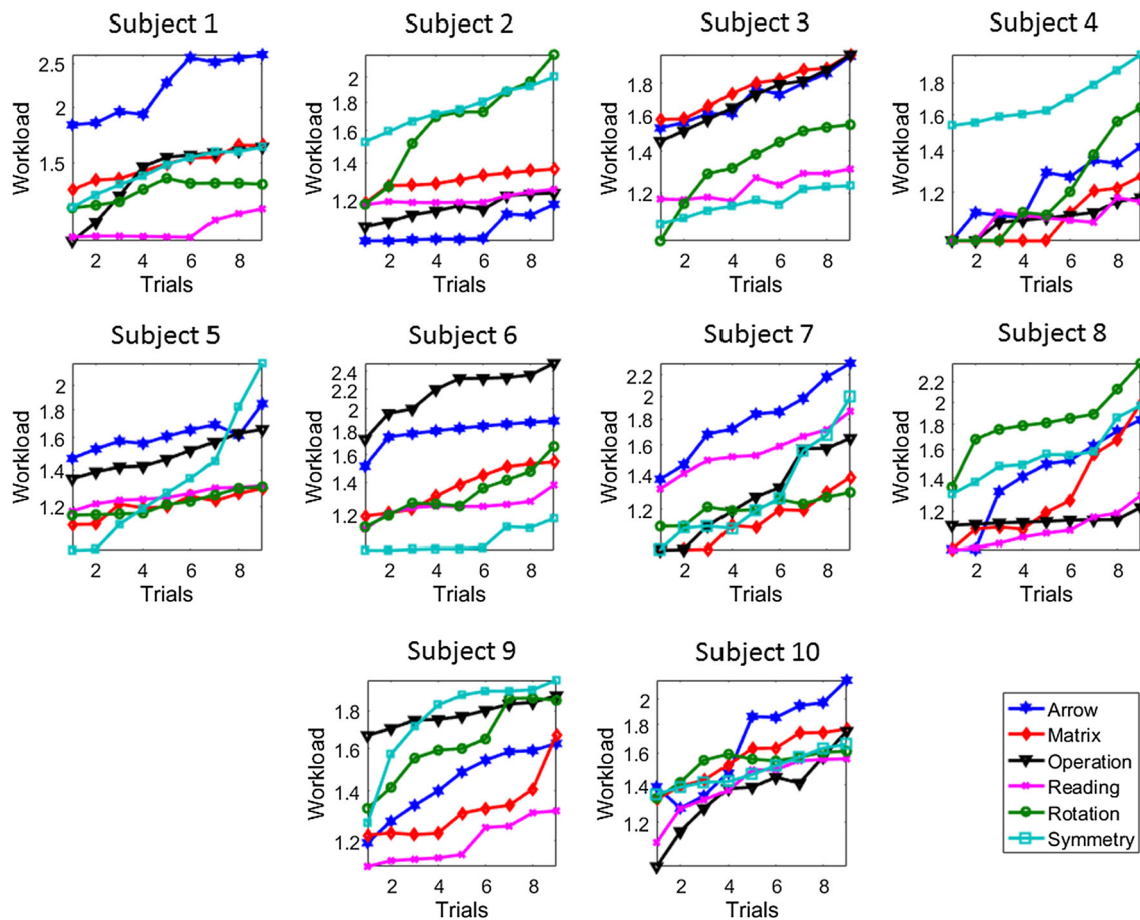


Fig. 13 Workload graph obtained for each participant in each task using the proposed approach

Table 3 Clustering coefficient of parietal area for alpha and beta bands

Level	Clustering coefficient of parietal area	
	Alpha	Beta
Low	0.594130 ± 0.039195	0.498129 ± 0.028613
Medium	0.593634 ± 0.036279	0.306449 ± 0.035995
High	0.581643 ± 0.038197	0.226450 ± 0.032712

2011; Sciaraffa et al. 2017). Besides, *local efficiency* shows decreasing trend for increase in workload from *low* to *high* (Zhang et al. 2017c). The results reveal that the trend is significant over the parietal, occipital, and temporal regions for the theta band but are not significant for the alpha band (Table 6). Figure 15 shows the number of connections at different load conditions; the results reveal that greater number of connections indicate less cognitive load (Fu et al. 2006; Huang et al. 2016; Mazher et al. 2017). Thus, it can be concluded that the proposed approach

successfully induces different load types and also has the potential for distinguishing the load data.

Comparison with related works

Although it is hard to perform a direct comparison of the state-of-the-art techniques due to the variability of EEG datasets, features utilized, participants and the tasks being performed. However, the proposed work has been compared with a few of the relevant literature as discussed in Sect. 1 on the basis of the mental task performed and the evaluation is carried on the obtained accuracy. The summary of the comparison study is given in Table 7.

From the table, it is evident that the estimation of workload using EEG has become quite prominent over a few years. Besides, EEG has also been combined with other modalities to improve classification accuracy. The majority of cognitive workload estimation work uses PSD features of EEG to classify workload. PSD has been explored in various ways to measure workload with accuracies varying between 85% to 93%. The proposed work

Table 4 Variation of *strength* in theta and alpha bands

Theta Rhythm			Alpha Rhythm		
Area	Level	Strength	Area	Level	Strength
Frontal	Low	6.699 ± 0.733	Frontal	Low	8.078 ± 0.862
	Med	6.916 ± 0.831		Med	8.601 ± 0.697
	High	8.048 ± 0.690		High	8.823 ± 0.783
Parietal	Low	7.757 ± 0.974	Parietal	Low	8.334 ± 0.975
	Med	7.797 ± 0.928		Med	8.629 ± 0.880
	High	8.073 ± 1.096		High	8.941 ± 0.622
Occipital	Low	7.640 ± 1.177	Occipital	Low	8.418 ± 1.126
	Med	7.750 ± 0.930		Med	8.687 ± 0.807
	High	7.890 ± 0.930		High	8.888 ± 0.830
Temporal	Low	7.220 ± 1.026	Temporal	Low	7.950 ± 1.048
	Med.	7.470 ± 0.922		Med	8.451 ± 0.910
	High	7.660 ± 0.590		High	8.605 ± 0.802

Table 5 One-way ANOVA measure *p*-value of theta and alpha bands for the parameter *strength*

Areas	Theta	Alpha
Frontal	0.963	0.662
Parietal	0.609	0.012*
Occipital	0.92	0.842
Temporal	0.798	0.033*

Symbol * indicates statistical difference between task difficulties ($p < 0.05$)

Table 6 One-way ANOVA measure *p*-value of theta and alpha bands for the parameter *local efficiency*

Areas	Theta	Alpha
Frontal	0.198	0.898
Parietal	0.005*	0.355
Occipital	0.053*	0.347
Temporal	0.049*	0.379

Symbol * indicates statistical difference between task difficulties ($p < 0.05$)

has greatly improved the classification of cognitive workload in different load conditions. Using a single layer ANN, the proposed approach has obtained a classification accuracy of 98.66%, which is better than most of the existing approaches.

Conclusion

Measuring cognitive workload has become an absolute necessity for various life risk jobs and critical industrial operations. Serious havoc may arise due to operator cognitive overload, which can lead to loss of human life and

other materialistic losses. The proposed work focuses on measuring compounded workload value for each individual. To meet the criteria, this work proposes a cognitive workload graph, which is dependent on the cognitive workload of the ‘former’ instance to measure the ‘current’ instance. Measuring workload in such a manner will remove any abruptness in the workload measurement. Further, using the graphical interpretation, the change in workload within an individual can be easily visualized. For designing the graph, this study has also introduced a workload defining vector comprising of ratio value of PSD of EEG rhythms using ‘idle’ and ‘experiment’ mental states. The defined feature vector can perfectly classify the

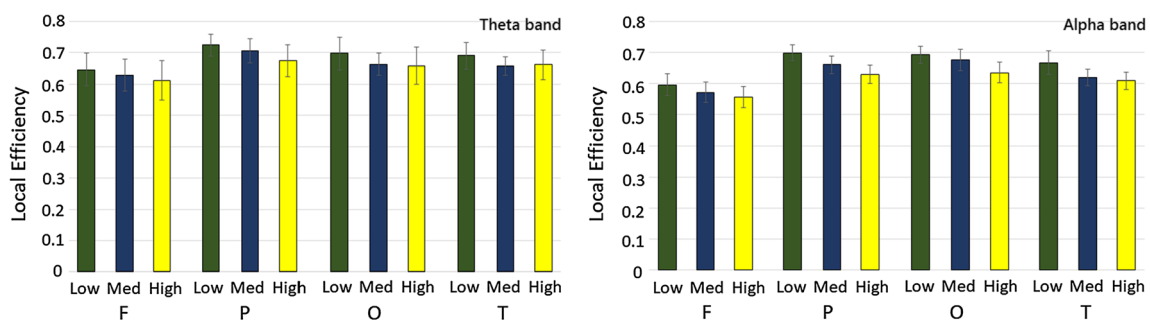
**Fig. 14** Variation of local efficiency for theta and alpha bands in each brain area and difficulty level

Fig. 15 Functional connectivity at each difficulty level

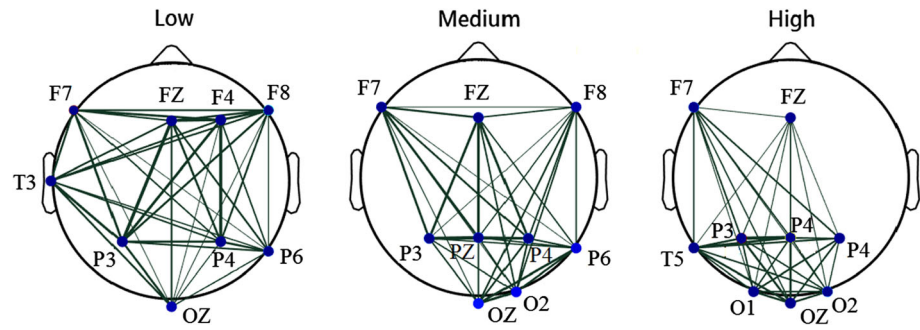


Table 7 Comparison of the proposed approach with state-of-the-art techniques

Author(s)	Cognitive task	Feature	Dataset: subject (data segments)	Classifier	Accuracy
Zhang et al. (2017b)	1-back, 2-back, and 3-back position memory tasks	EEG (PSD and ERP features) and ECG (HRV feature)	10 (6000)	k-NN, SVM	IMIM-F 91% IMIM-C 90%
Zhao et al. (2018)	Verbal 1-, 3-, and 4-back tasks	SSVEPs of EEG	15 (3600)	-	93.91% - 96.09%
Mühl et al. (2014)	n-back task	ERP and CSP of EEG	24 (1080)	LDA	76.1% - 96%
Ke et al. (2014)	Verbal n-back task (n = 0,1,2,3)	Sums of PSD in EEG frequency bands	17 (153)	SVM Regressor	-
Radüntz (2017)	0-back, 2-back	EEG- Dual Frequency Head Maps	54 (486)	SVM	Low load 94.06% High load 92.69%
Yu et al. (2015)	n-back task based on visual degradation	PSD and BCSP from ERPs of EEG	16 (1920)	Linear probabilistic LIBSVM	85%
Aghajani et al. (2017)	Letter n-back task (0, 1, 2 and 3)	EEG features, fNIRS features and EEG+fNIRS features	14 (560)	linear support vector machine (SVM)	Binary: EEG 85.9% fNIRS 74.8% EEG+fNIRS 90.9% Multi-class: EEG 79.6% fNIRS 57.0% EEG+fNIRS 86.2%
Mahmoud et al. (2017)	n-back task consisting of digits and images.	spectral power density, statistical features, morphological features, and time-frequency features of EEG	16 (1920)	Linear probabilistic LIBSVM	93.4%
Proposed approach	n-back tasks	vector comprising of ratio of PSD of EEG rhythms	10 (540)	ANN	98.66%

load states with an accuracy of 98.66% using an ANN classifier. To validate the induction of workload into the EEG signals, this study uses functional connectivity parameters which successfully affirm that the collected raw EEG data has load of different difficulty levels. Future work will focus on using the workload metrics to design human-machine interaction systems that constantly monitor workload imposed on the operators and help in task allocation based on the operator's mental state.

Data availability The data will be available on request to the corresponding author.

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